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Arthroplasty implants, systems, and methods

Abstract

A bone implant may include a shaft having a proximal end, a distal end, a longitudinal axis, a proximal shaft portion, and a distal shaft portion. The proximal shaft portion may include a first minor diameter, and a first helical thread disposed about the proximal shaft portion defining a first major diameter. The first helical thread may include a first concave undercut surface. The distal shaft portion may include a second minor diameter, and a second helical thread disposed about the distal shaft portion defining a second major diameter. The second helical thread may include a second concave undercut surface. The first and second concave undercut surfaces may be angled towards the distal end of the shaft. The second minor diameter may be smaller than the first minor diameter and the second major diameter may be smaller than the first major diameter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/147,640 filed on Feb. 9, 2021, entitled “FASTENING DEVICES, SYSTEMS, AND METHODS”. The foregoing document is hereby incorporated by reference in its entirety.

TECHNICAL FIELD
(1) The present disclosure relates to joint arthroplasty implants. More specifically, the present disclosure relates to shoulder arthroplasty implants with improved thread designs and morphology. While the present disclosure is made in the context of humeral and glenoid implants for shoulder arthroplasty, the disclosed principles are applicable to arthroplasty implants for other locations.
BACKGROUND
(2) Joint arthroplasty procedures are conducted to restore the function of an unhealthy joint. Typically, these procedures involve replacing the unhealthy natural articulating surfaces of the joint with artificial articulating surfaces. The new artificial articulating surfaces are typically anchored into the adjacent bones to maintain long term stability. However, joint arthroplasty devices implanted within bone can become lose over time due to multi-axial forces and off-axis loading scenarios that may be applied to the joint arthroplasty device during the healing process. Joint arthroplasty devices utilizing traditional thread designs, tapered stems, keels, or other methods may not provide sufficient fixation to overcome these multi-axial forces and off-axis loading scenarios. Accordingly, joint arthroplasty devices with improved thread designs for increasing bone fixation and load sharing between a bone/implant interface experiencing multi-axial and off-loading conditions would be desirable.
(3) In shoulder arthroplasty, a humeral implant is attached to the humerus, and a glenoid implant is attached to the glenoid or scapula. There are two different main categories of shoulder arthroplasty: anatomic and reverse. In an anatomic procedure, the implant designs are intended to replicate the natural anatomy. The humeral head is replaced with a similarly shaped convex hemispherical surface, while the glenoid is replaced with a shallow concave socket. In a reverse procedure, the natural ball and socket is reversed. The humeral head is replaced with a socket fixed to the humerus and the glenoid is replaced with a ball (or glenosphere) fixed to the scapula.
(4) Regardless of the type of procedure, fixation of the humeral component into the humerus typically involves an implant with a shaft portion that extends into the metaphysis and optionally into the diaphysis of the humerus. The goals of these implants are to preserve as much native bone as possible, maximize the mechanical stability of the implants, and allow for more physiological loading of the bone to preserve long-term fixation.
(5) The present disclosure is made in the context of humeral and glenoid implants for shoulder arthroplasty. Other applications may include a femoral implant for hip or knee arthroplasty, a tibial implant for knee or ankle arthroplasty, implants for the elbow, wrist, hand, foot, etc.
SUMMARY
(6) The various arthroplasty implants, systems, and methods of the present disclosure have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available arthroplasty implants, systems, and methods. In some embodiments, the arthroplasty implants, systems, and methods of the present disclosure may provide improved bone fixation and load sharing between a

bone/implant interface under multi-axial and off-loading conditions.

(7) In some embodiments, a bone implant may include a shaft including a proximal end, a distal end, a longitudinal axis, a proximal shaft portion, and a distal shaft portion. The proximal shaft portion may include a first minor diameter and a first helical thread disposed about the proximal shaft portion defining a first major diameter of the proximal shaft portion. The first helical thread may include a first concave undercut surface. The distal shaft portion may include a second minor diameter and a second helical thread disposed about the distal shaft portion defining a second major diameter of the distal shaft portion. The second helical thread may include a second concave undercut surface. The first concave undercut surface and the second concave undercut surface may be angled towards the distal end of the shaft, the second minor diameter of the distal shaft portion may be smaller than the first minor diameter of the proximal shaft portion, and the second major diameter of the distal shaft portion may be smaller than the first major diameter of the proximal shaft portion.

(8) In some embodiments, the bone implant may include a flange component at the proximal end of the shaft. The flange component may include a bone-facing surface and an implement-facing surface.

(9) In some embodiments, the bone-facing surface may include a convex surface.

(10) In some embodiments, the bone-facing surface may include a semi-spherical surface.

(11) In some embodiments, the bone implant may include an attachment feature at the proximal end of the shaft configured to secure an implement.

(12) In some embodiments, the attachment feature may include a post, and the implement may include an articulating head comprising a convex semi-spherical articular surface. The articulating head may be configured to be removably couplable with the post.

(13) In some embodiments, the attachment feature may include a recess, and the implement may include an insert having a concave semi-spherical articular surface. The insert may be configured to be removably couplable with the recess.

(14) In some embodiments, a bone implant may include a shaft, an articulating member disposed at the proximal end of the shaft, and a helical thread. The shaft may include a proximal end, a distal end, a longitudinal axis, a minor diameter, and a threaded shaft portion. The helical thread may be disposed about the shaft defining a length of the threaded shaft portion. The helical thread may include a concave undercut surface angled towards the distal end of the shaft, and a ratio of the length of the threaded shaft portion to the minor diameter of the shaft may be less than 1.50.

(15) In some embodiments, the ratio of the length of the threaded shaft portion to the minor diameter of the shaft may be less than 1.25.

(16) In some embodiments, the ratio of the length of the threaded shaft portion to the minor diameter of the shaft may be less than 1.10.

(17) In some embodiments, the ratio of the length of the threaded shaft portion to the minor diameter of the shaft may be equal to 1.0.

(18) In some embodiments, the ratio of the length of the threaded shaft portion to the minor diameter of the shaft may be less than 1.0.

(19) In some embodiments, the bone implant may include a flange component at the proximal end of the shaft. The flange component may include a bone-facing surface and an implement-facing surface.

(20) In some embodiments, the articulating member may be configured to be removably couplable with an attachment feature at the proximal end of the shaft.

(21) In some embodiments, a shoulder joint implant may include a shaft, a helical thread, and a shoulder joint implement comprising an articular surface at the proximal end of the shaft. The shaft may include a proximal end, a distal end, and a longitudinal axis. The helical thread may be disposed about the shaft along the longitudinal axis between the proximal and distal ends of the shaft. The helical thread may include a first undercut surface, a second undercut surface, a third

undercut surface, and a fourth open surface. The first undercut surface and the third undercut surface may be angled towards one of the proximal end and the distal end of the shaft. The second undercut surface and the fourth open surface may be angled towards the other one of the proximal end and the distal end of the shaft.
(22) In some embodiments, the shoulder joint implement may include a glenoid head prosthesis, and the articular surface may include a convex semi-spherical surface.
(23) In some embodiments, the shoulder joint implement may include a humeral head prosthesis, and the articular surface may include a convex semi-spherical surface.
(24) In some embodiments, the shoulder joint implement may include a glenoid insert prosthesis, and the articular surface may include a concave semi-spherical surface.
(25) In some embodiments, the shoulder joint implement may include a humeral insert prosthesis, and the articular surface may include a concave semi-spherical surface.
(26) In some embodiments, the shoulder joint implant may include a flange component at the proximal end of the shaft. The flange component may include a bone-facing surface and an implement-facing surface.
(27) These and other features and advantages of the present disclosure will become more fully apparent from the following description and appended claims or may be learned by the practice of the devices, systems, and methods set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Exemplary embodiments of the present disclosure will become more fully apparent from the following description taken in conjunction with the accompanying drawings. Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the scope of the present disclosure, the exemplary embodiments of the present disclosure will be described with additional specificity and detail through use of the accompanying drawings in which:
(2) FIG. **1**A illustrates a front perspective view of a fastener, according to an embodiment of the present disclosure; FIG. **1**B illustrates a rear perspective view of the fastener of FIG. **1**A; FIG. **1**C illustrates a side view of the fastener of FIG. **1**A; FIG. **1**D illustrates a cross-sectional side view of the fastener of FIG. **1**A taken along the line A-A shown in FIG. **1**C;
(3) FIG. **2** illustrates a partial cross-sectional side view of a fastener comprising crescent-shaped threading;
(4) FIG. **3**A illustrates a front perspective view of a bone implant, according to an embodiment of the present disclosure; FIG. **3**B illustrates a rear perspective view of the bone implant of FIG. **3**A; FIG. **3**C illustrates a side view of the bone implant of FIG. **3**A; FIG. **3**D illustrates a cross-sectional side view of the bone implant of FIG. **3**A taken along the line B-B shown in FIG. **3**C;
(5) FIG. **4**A illustrates a front perspective view of a bone implant, according to another embodiment of the present disclosure; FIG. **4**B illustrates a rear perspective view of the bone implant of FIG. **4**A; FIG. **4**C illustrates a side view of the bone implant of FIG. **4**A; FIG. **4**D illustrates a cross-sectional side view of the bone implant of FIG. **4**A taken along the line C-C shown in FIG. **4**C;
(6) FIG. **5**A illustrates a front perspective view of a bone implant, according to another embodiment of the present disclosure; FIG. **5**B illustrates a rear perspective view of the bone implant of FIG. **5**A; FIG. **5**C illustrates a bottom view of the bone implant of FIG. **5**A; FIG. **5**D illustrates a top view of the bone implant of FIG. **5**A; FIG. **5**E illustrates a side view of the bone implant of FIG. **5**A;
(7) FIG. **6**A illustrates a perspective view of an implement, according to an embodiment of the

present disclosure; FIG. **6**B illustrates a side view of the implement of FIG. **6**A;
(8) FIG. **7**A illustrates a bottom perspective view of an implement, according to another embodiment of the present disclosure; FIG. **7**B illustrates a top perspective view of the implement of FIG. **7**A;
(9) FIG. **8** illustrates a front view of a shoulder joint, according to an embodiment of the present disclosure;
(10) FIG. **9** illustrates a perspective view of a scapula bone of the shoulder joint of FIG. **8**;
(11) FIG. **10** illustrates the scapula bone of FIG. **9** with a prepared glenoid;
(12) FIG. **11** illustrates a side view of the scapula bone of FIG. **10**;
(13) FIG. **12** illustrates a cross-sectional side view of the scapula bone of FIG. **11** taken along the line D-D shown in FIG. **11**;
(14) FIG. **13** illustrates a perspective view of a humeral bone of the shoulder joint of FIG. **8**;
(15) FIG. **14** illustrates the humeral bone of FIG. **13** with a prepared humeral head;
(16) FIG. **15** illustrates a side view of the humeral bone of FIG. **14**;
(17) FIG. **16** illustrates a cross-sectional side view of the humeral bone of FIG. **15**;
(18) FIG. **17** illustrates a front view of the shoulder joint of FIG. **8** with prepared humeral and scapula bones prior to receiving one or more shoulder joint implants;
(19) FIG. **18** illustrates a front view of the shoulder joint of FIG. **17** with a reverse shoulder arthroplasty system;
(20) FIG. **19** illustrates a cross-sectional side view of the shoulder joint shown in FIG. **18**;
(21) FIG. **20** illustrates a front perspective view of a reverse shoulder arthroplasty system, according to an embodiment of the present disclosure;
(22) FIG. **21** illustrates a front perspective view of an anatomic shoulder arthroplasty system, according to an embodiment of the present disclosure;
(23) FIG. **22**A illustrates a front perspective view of a bone implant, according to another embodiment of the present disclosure; FIG. **22**B illustrates a rear perspective view of the bone implant of FIG. **22**A; FIG. **22**C illustrates a side view of the bone implant of FIG. **22**A; and FIG. **22**D illustrates a cross-sectional side view of the bone implant of FIG. **22**A taken along the line E-E shown in FIG. **22**C.
(24) It is to be understood that the drawings are for purposes of illustrating the concepts of the present disclosure and may not be drawn to scale. Furthermore, the drawings illustrate exemplary embodiments and do not represent limitations to the scope of the present disclosure.

DETAILED DESCRIPTION

(25) Exemplary embodiments of the present disclosure will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood that the components of the present disclosure, as generally described and illustrated in the drawings, could be arranged, and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the implants, systems, and methods, as represented in the drawings, is not intended to limit the scope of the present disclosure but is merely representative of exemplary embodiments of the present disclosure.
(26) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in the drawings, the drawings are not necessarily drawn to scale unless specifically indicated.
(27) FIGS. **1**A-**1**D illustrate various views of a fastener **100**, implantable bone anchor, or bone screw, according to one embodiment of the present disclosure. Specifically, FIG. **1**A is a front perspective view of the fastener **100**, FIG. **1**B is a rear perspective view of the fastener **100**, FIG. **1**C is a side view of the fastener **100**, and FIG. **1**D is a cross-sectional side view of the fastener **100** taken along the line A-A in FIG. **1**C.

(28) In general, the fastener **100** may include a shaft **105** having a proximal end **101**, a distal end **102**, and a longitudinal axis **103**. The fastener **100** may also include a head **104** located at the proximal end **101** of the shaft **105**, a torque connection interface **106** formed in/on the head **104** (in either a male/female configuration), and a self-tapping feature **107** formed in the distal end **102** of the shaft **105**.
(29) In some embodiments, the fastener **100** may include a first helical thread **110** disposed about the shaft **105**, and a second helical thread **120** disposed about the shaft **105** adjacent the first helical thread **110**.
(30) In some embodiments, the fastener **100** may include a “dual start” or “dual lead” thread configuration comprising the first helical thread **110** and the second helical thread **120**.
(31) In some embodiments, a depth of the first helical thread **110** and/or the second helical thread **120** with respect to the shaft **105** may define a major diameter vs. a minor diameter of the shaft **105** alone.
(32) In some embodiments, a major diameter and/or a minor diameter of the fastener **100** may be constant or substantially constant along the entire length of the fastener, or along a majority of the length of the fastener. In these embodiments, a constant minor diameter may help avoid blowout of narrow/delicate bones (e.g., a pedicle) when inserting a fastener into a bone. In some embodiments, a pilot hole may first be drilled into a narrow/delicate bone and then a fastener having a similar minor diameter in comparison to the diameter of the pilot hole may be chosen to avoid blowout when inserting the fastener into the bone, as will be discussed in more detail below.
(33) In some embodiments, a depth of the first helical thread **110** and/or the second helical thread **120** with respect to the shaft **105** may vary along a length of the shaft **105** to define one or more major diameters of the fastener **100** and/or one or more regions along the fastener **100** may comprise one or more continuously variable major diameters.
(34) In some embodiments, a thickness of the shaft **105** may vary along a length of the shaft **105** to define one or more minor diameters of the fastener **100**, and/or one or more regions along the fastener **100** may comprise one or more continuously variable minor diameters. In some embodiments, a thickness/height/width/length/pitch/angle/shape, etc., of the first helical thread **110** and/or the second helical thread **120** (or any additional helical thread) may vary along a length of the shaft **105**. For example, a thickness/height/width/length/pitch/angle/shape, etc., of the first helical thread **110** and/or the second helical thread **120** may be greater towards the tip of the fastener and thinner towards the head of the fastener (or vice versa) in either a discrete or continuously variable fashion, etc.
(35) In some embodiments, the major and/or minor diameters may increase toward a proximal end or head of a fastener in order to increase bone compaction as the fastener is terminally inserted into the bone/tissue.
(36) In some embodiments, a pitch of the first helical thread **110** and/or the second helical thread **120** may vary along a length of the fastener **100**.
(37) In some embodiments, the fastener **100** may include a plurality of helical threads disposed about the shaft **105**. However, it will also be understood that any of the fasteners disclosed or contemplated herein may include a single helical thread disposed about the shaft of the fastener. Moreover, the fastener **100** may comprise a nested plurality of helical threads having different lengths (not shown). As one non-limiting example, the fastener **100** may include a first helical thread **110** that is longer than a second helical thread **120**, such that the fastener **100** comprises dual threading along a first portion of the shaft **105** and single threading along a second portion of the shaft **105**.
(38) In some embodiments, the plurality of helical threads may include three helical threads comprising a “triple start” or “triple lead” thread configuration (not shown).
(39) In some embodiments, the plurality of helical threads may include four helical threads comprising a “quadruple start” or “quadruple lead” thread configuration (not shown).

(40) In some embodiments, the plurality of helical threads may include more than four helical threads (not shown).
(41) In some embodiments, the fastener **100** may include first threading with any of the shapes disclosed herein oriented toward one of the proximal end and the distal end of the fastener **100**, with the first threading located proximate the distal end of the fastener **100**, as well as second threading with any of the shapes disclosed herein oriented toward the other one of the proximal end and the distal end of the fastener **100**, with the second threading located proximate the head of the fastener **100** (not shown).
(42) In some embodiments, the fastener **100** may include multiple threading (e.g., dual helical threading, etc.) with any of the shapes disclosed herein located proximate one of the proximal end and the distal end of the fastener **100**, as well as single threading with any of the shapes disclosed herein with the second threading located proximate the other of the proximal end and the distal end of the fastener **100** (not shown).
(43) In some embodiments, the first helical thread **110** may include a plurality of first concave undercut surfaces **131** and a plurality of first convex undercut surfaces **141**.
(44) In some embodiments, the second helical thread **120** may include a plurality of second concave undercut surfaces **132** and a plurality of second convex undercut surfaces **142**.
(45) In some embodiments, when the fastener **100** is viewed in section along a plane that intersects the longitudinal axis **103** of the shaft **105** (e.g., see FIG. **1**D), the plurality of first concave undercut surfaces **131** and the plurality of second convex undercut surfaces **142** may be oriented toward (i.e., point toward) the proximal end **101** of the shaft **105**.
(46) In some embodiments, the plurality of first convex undercut surfaces **141** and the plurality of second concave undercut surfaces **132** may be oriented toward (i.e., point toward) the distal end **102** of the shaft **105**.
(47) In some embodiments, at least one of the plurality of first concave undercut surfaces **131**, the plurality of first convex undercut surfaces **141**, the plurality of second concave undercut surfaces **132**, and the plurality of second convex undercut surfaces **142** may comprise at least one substantially flat surface.
(48) In some embodiments, when the fastener **100** is viewed in section along a plane intersecting the longitudinal axis **103** of the shaft **105**, the first helical thread **110** may comprise a plurality of first bent shapes (comprising at least one surface that is angled relative to the longitudinal axis **103** of the shaft **105** and/or at least one undercut surface) with a plurality of first intermediate portions **151** that are oriented toward (i.e., point toward) the distal end **102** of the shaft **105**. This may be referred to as “standard” threading, having a “standard” orientation.
(49) In some embodiments, when the fastener **100** is viewed in section along a plane intersecting the longitudinal axis **103** of the shaft **105**, the second helical thread **120** may comprise a plurality of second bent shapes (comprising at least one surface that is angled relative to the longitudinal axis **103** of the shaft **105** and/or at least one undercut surface) with a plurality of second intermediate portions **152** that are oriented toward (i.e., point toward) the proximal end **101** of the shaft **105**. This may be referred to as “inverted” threading, having an “inverted” orientation.
(50) In some embodiments, one or more helical threads may morph/transition between a standard orientation and an inverted orientation along a shaft of a fastener.
(51) In some embodiments, at least one of the plurality of first concave undercut surfaces **131**, the plurality of first convex undercut surfaces **141**, the plurality of second concave undercut surfaces **132**, and the plurality of second convex undercut surfaces **142** may comprise at least one curved surface.
(52) As shown in FIG. **1**D, the proximally-oriented and distally-oriented surfaces of the first helical thread **110** (i.e., the first concave undercut surfaces **131** and the first convex undercut surfaces **141** in the fastener **100** of FIG. **1**D) may not have mirror symmetry relative to each other about any plane perpendicular to the longitudinal axis **103** of the fastener **100**. Rather, the first concave

undercut surfaces **131** and the first convex undercut surfaces **141** may be generally parallel to each other. The same may be true for the second helical thread **120**, in which the second concave undercut surfaces **132** and the second convex undercut surfaces **142** may not have mirror symmetry relative to each other but may be generally parallel to each other.
(53) Conversely, as also shown in FIG. **1**D, the proximally-oriented surfaces of the first helical thread **110** may have mirror symmetry relative to the distally-oriented surfaces of the second helical thread **120**. Specifically, the first concave undercut surfaces **131** may have mirror symmetry relative to the second convex undercut surfaces **142** about a plane **170** that bisects the space between them and lies perpendicular to the longitudinal axis **103**.
(54) Similarly, the distally-oriented surfaces of the first helical thread **110** may have mirror symmetry relative to the proximally-oriented surfaces of the second helical thread **120**. Specifically, the second concave undercut surfaces **132** may have mirror symmetry relative to the first convex undercut surfaces **141** about a plane **172** that bisects the space between them and lies perpendicular to the longitudinal axis **103**.
(55) This mirror symmetry may be present along most of the length of the first helical thread **110** and the second helical thread **120**, with symmetry across different planes arranged between adjacent turns of the first helical thread **110** and the second helical thread **120** along the length of the longitudinal axis **103**. Such mirror symmetry may help more effectively capture bone between the first helical thread **110** and the second helical thread **120** and may also facilitate manufacture of the fastener **100**.
(56) In some embodiments, when the fastener **100** is viewed in section along a plane intersecting the longitudinal axis **103** of the shaft **105**, the first helical thread **110** may include at least one partial crescent shape that is oriented toward (i.e., points toward) the distal end **102** of the shaft **105** and/or the proximal end **101** of the shaft **105**. FIG. **2** illustrates a partial cross-sectional view of a fastener **200** comprising crescent shapes, as one non-limiting example of such an embodiment.
(57) In some embodiments (not shown), when the fastener **100** is viewed in section along a plane intersecting the longitudinal axis **103** of the shaft **105**, the first helical thread **110** may include at least one partial crescent shape that is oriented toward (i.e., points toward) the distal end **102** of the shaft **105**, and the second helical thread **120** may include at least one partial crescent shape that is oriented toward (i.e., points toward) the proximal end **101** of the shaft **105**.
(58) In some embodiments (not shown), the first helical thread **110** may include a first plurality of partial crescent shapes that are oriented toward (i.e., point toward) the distal end **102** of the shaft **105**, and the second helical thread **120** may include a second plurality of partial crescent shapes that are oriented toward (i.e., point toward) the proximal end **101** of the shaft **105**.
(59) In some embodiments (not shown), the first plurality of partial crescent shapes and the second plurality of partial crescent shapes may be arranged in alternating succession along the shaft **105** of the fastener **100**.
(60) In some embodiments, the first helical thread **110** may be bisected by the line **123** shown in FIG. **2** with each crescent shape including a plurality of first undercut surfaces **111**, a plurality of second undercut surfaces **112**, a plurality of third undercut surfaces **113**, and a plurality of fourth open surfaces **114** similar to the helical threading shown in FIG. **1**D, except with curved surfaces in place of flat surfaces.
(61) In some embodiments, the plurality of first undercut surfaces **111** and the plurality of second undercut surfaces **112** may comprise concave curved surfaces. However, it will be understood that portions of the plurality of first undercut surfaces **111** and/or portions of the plurality of second undercut surfaces **112** may also comprise convex curved surfaces and/or flat surfaces (not shown in FIG. **2**).
(62) In some embodiments, the plurality of third undercut surfaces **113** and the plurality of fourth open surfaces **114** may comprise convex curved surfaces. However, it will be understood that portions of the plurality of third undercut surfaces **113** and the plurality of fourth open surfaces **114**

may also comprise concave curved surfaces and/or flat surfaces (not shown in FIG. **2**).
(63) In some embodiments, the plurality of third undercut surfaces **113** and the plurality of fourth open surfaces **114** may be replaced by a ramped surface (such as that utilized in a standard buttress thread design) without any undercuts (not shown in FIG. **2**). Likewise, any of the other thread designs disclosed herein may utilize a ramped or buttress thread design on at least one side of the helical thread.
(64) In some embodiments, the plurality of first undercut surfaces **111**, the plurality of second undercut surfaces **112**, the plurality of third undercut surfaces **113**, and/or the plurality of fourth open surfaces **114** may comprise a mixture of curved surfaces and flat/faceted surfaces (not shown in FIG. **2**). For example, in some embodiments the plurality of first undercut surfaces **111** and the plurality of second undercut surfaces **112** may comprise concave curved surfaces forming a crescent shape, and the plurality of third undercut surfaces **113** and the plurality of fourth open surfaces **114** may comprise flat/faceted surfaces forming a convex undercut surface, similar to the first convex undercut surfaces **141** shown in FIG. **1**D.
(65) In some embodiments, a fastener may have only standard threads or only inverted threads. The type of threads that are desired may depend on the type and/or magnitude of loads to be applied to the fastener. For example, a screw loaded axially away from the bone in which it is implanted may advantageously have a standard thread, while a screw loaded axially toward the bone in which it is implanted may advantageously have an inverted thread. A screw that may experience multi-axial loading and/or off-loading conditions may advantageously include at least one standard thread and at least one inverted thread in order to increase bone fixation and load sharing between a bone/fastener interface during multi-axial and off-loading conditions to reduce high bone strain and distribute multi-axial forces applied to the bone in a load-sharing, rather than load-bearing, configuration. Shear loads and/or bending moments may also be optimally resisted with any chosen combination of threading, threading morphology, and/or threading variations contemplated herein to optimally resist shear loads, bending moments, multi-axial loading, off-loading conditions, etc.
(66) In some embodiments, fasteners with standard threads may be used in conjunction with fasteners with inverted threads in order to accommodate different loading patterns.
(67) In some embodiments, a single fastener may have both standard and inverted threads, like the fastener **100**. Such a combination of threads may help the fastener **100** remain in place with unknown and/or varying loading patterns.
(68) In some embodiments, the geometry of the threading of a fastener (with standard and/or inverted threads) may be varied to suit the fastener for a particular loading scheme. For example, the number of threads, the number of thread starts, the pitch of the threading, the lead(s) of the threading, the shape(s) of the threading, any dimension(s) associated with the threading (e.g., any length(s)/width(s)/height(s), etc., associated with the threading), the major diameter(s), the minor diameter(s), any angulation/angles associated with any surfaces of the threading, the “handedness” of the threading (e.g., right-handed vs. left-handed), etc., may be varied accordingly to suit any specific medium of installation, loading pattern, desired radial loading force, pull-out strength, application, procedure, etc., that may be involved.
(69) In some embodiments, the material(s) of any portion of a fastener described herein may include, but are not limited to metals (e.g., titanium, cobalt, stainless steel, etc.), metal alloys, plastics, polymers, PEEK, UHMWPE, composites, additive particles, textured surfaces, biologics, biomaterials, bone, etc.
(70) In some embodiments, any of the fasteners described herein may include additional features such as: self-tapping features, locking features (e.g., locking threading formed on a portion of the fastener, such as threading located on or near a head of the fastener), cannulation, any style of fastener head (or no fastener head at all), any style of torque connection interface (or no torque connection interface at all), etc.
(71) In some embodiments, a tap (not shown) may be utilized to pre-form threading in a bone

according to any threading shape that is disclosed or contemplated herein. In this manner, taps with any suitable shape may be utilized in conjunction with any fastener described or contemplated herein to match or substantially match the threading geometry of a given fastener.
(72) In some embodiments, a minor diameter of the fastener may be selected to match, or substantially match, a diameter of a pilot hole that is formed in a bone to avoid bone blowout when the fastener is inserted into the pilot hole, as will be discussed in more detail below.
(73) Additionally, or alternatively thereto, the type of threads and/or thread geometry may be varied based on the type of bone in which the fastener is to be anchored. For example, fasteners anchored in osteoporotic bone may fare better with standard or inverted threads, or when the pitch, major diameter, and/or minor diameter are increased or decreased, or when the angulation of thread surfaces is adjusted, etc.
(74) In some embodiments, a surgical kit may include multiple fasteners with any of the different fasteners and thread options described or contemplated herein. The surgeon may select the appropriate fastener(s) from the kit based on the particular loads to be applied and/or the quality of bone in which the fastener(s) are to be anchored.
(75) Continuing with FIG. **1**D, in some embodiments the first helical thread **110** may include a plurality of first undercut surfaces **111**, a plurality of second undercut surfaces **112**, a plurality of third undercut surfaces **113**, and a plurality of fourth open surfaces **114**.
(76) In some embodiments, the second helical thread **120** may include a plurality of fifth undercut surfaces **125**, a plurality of sixth undercut surfaces **126**, a plurality of seventh undercut surfaces **127**, and a plurality of eighth open surfaces **128**.
(77) In some embodiments one or more of the plurality of first undercut surfaces **111**, the plurality of second undercut surfaces **112**, the plurality of third undercut surfaces **113**, the plurality of fourth open surfaces **114**, the plurality of fifth undercut surfaces **125**, the plurality of sixth undercut surfaces **126**, the plurality of seventh undercut surfaces **127**, and the plurality of eighth open surfaces **128** may comprise at least one flat or substantially flat surface.
(78) In some embodiments, the plurality of first undercut surfaces **111**, the plurality of third undercut surfaces **113**, the plurality of sixth undercut surfaces **126**, and the plurality of eighth open surfaces **128** may be angled towards the distal end **102** of the shaft **105**.
(79) In some embodiments, the plurality of second undercut surfaces **112**, the plurality of fourth open surfaces **114**, the plurality of fifth undercut surfaces **125**, and the plurality of seventh undercut surfaces **127** may be angled towards the proximal end **101** of the shaft **105**.
(80) In some embodiments, when the fastener **100** is viewed in section along a plane that intersects the longitudinal axis **103** of the shaft **105** (as shown in FIG. **1**D), the first helical thread **110** may include at least one chevron shape that is oriented toward (i.e., points toward) the distal end **102** of the shaft **105**. Likewise, the second helical thread **120** may also include at least one chevron shape that is oriented toward (i.e., points toward) the proximal end **101** of the shaft **105**.
(81) In some embodiments, when the fastener **100** is viewed in section along a plane that intersects the longitudinal axis **103** of the shaft **105** (as shown in FIG. **1**D), the first helical thread **110** may include a first plurality of chevron shapes that are oriented toward (i.e., point toward) the distal end **102** of the shaft **105**. Likewise, the second helical thread **120** may include a second plurality of chevron shapes that are oriented toward (i.e., point toward) the proximal end **101** of the shaft **105**.
(82) In some embodiments, the first plurality of chevron shapes and the second plurality of chevron shapes may be arranged in alternating succession along the shaft **105** of the fastener **100**, (e.g., see FIG. **1**D).
(83) In some embodiments, a plurality of first interlocking spaces **161** and a plurality of second interlocking spaces **162** may be formed between the first helical thread **110** and the second helical thread **120** along the shaft **105** of the fastener **100**.
(84) In some embodiments, the plurality of first interlocking spaces **161** may be formed intermediate the first concave undercut surfaces **131** and the second concave undercut surfaces **132**.

(85) In some embodiments, the plurality of second interlocking spaces **162** may be formed intermediate the first convex undercut surfaces **141** and the second convex undercut surfaces **142**.
(86) In some embodiments, the plurality of first interlocking spaces **161** may be larger in size than the plurality of second interlocking spaces.
(87) In some embodiments, the plurality of first interlocking spaces **161** and the plurality of second interlocking spaces **162** may be shaped and/or configured to interlock with bone/other tissues received therein to increase fixation of the fastener **100** within the bone/other tissues and provide additional resistance against multi-axial forces that may be applied to the fastener **100** and/or the bone/other tissues.
(88) In some embodiments, the plurality of second undercut surfaces **112** and the plurality of sixth undercut surfaces **126** may be angled toward each other to trap bone/other tissues within the plurality of first interlocking spaces **161** in order to increase fixation and resistance against multi-axial forces.
(89) In some embodiments, the plurality of third undercut surfaces **113** and the plurality of seventh undercut surfaces **127** may be angled toward each other to trap bone/other tissues within the plurality of second interlocking spaces **162** in order to increase fixation and resistance against multi-axial forces.
(90) In some embodiments, the plurality of first undercut surfaces **111** and the plurality of fifth undercut surfaces **125** may each form an angle α with respect to the longitudinal axis **103** of the shaft **105**, as shown in FIG. **1**D.
(91) In some embodiments, the angle α may be greater than 90 degrees.
(92) In some embodiments, the plurality of second undercut surfaces **112** and the plurality of sixth undercut surfaces **126** may each form an angle β with respect to the longitudinal axis **103** of the shaft **105**.
(93) In some embodiments, the angle β may be less than 90 degrees.
(94) In some embodiments, the plurality of third undercut surfaces **113** and the plurality of seventh undercut surfaces **127** may each form an angle θ with respect to the longitudinal axis **103** of the shaft **105**.
(95) In some embodiments, the angle θ may be approximately 90 degrees.
(96) In some embodiments, the angle θ may be greater than 90 degrees.
(97) It will be understood that the fastener **100** may include any thread configuration, feature, or morphology described or contemplated herein to achieve optimal fixation within a given bone/tissue. Moreover, it will also be understood that the fastener **100** may be utilized in conjunction with (or within) any system, method, or instrumentation described or contemplated herein.
(98) FIGS. **3**A-**3**D illustrate various views of a fastener, implant, shoulder joint implant, or bone implant **300**, according to another embodiment of the present disclosure. Specifically, FIG. **3**A is a front perspective view of the bone implant **300**, FIG. **3**B is a rear perspective view of the bone implant **300**, FIG. **3**C is a side view of the bone implant **300**, and FIG. **3**D is a cross-sectional side view of the bone implant **300** taken along the line B-B in FIG. **3**C.
(99) The bone implant **300** may generally include a shaft **305** having a proximal end **301**, a distal end **302**, a longitudinal axis **303**, a proximal shaft portion **308**, a distal shaft portion **309**, a first helical thread **310**, a second helical thread **320**, one or more self-tapping features **307**, a flange component **380**, an attachment feature or implant post **390**, and a torque connection interface **306** formed in/on the implant post **390**.
(100) In some embodiments, the proximal shaft portion **308** may have a first minor diameter **321** and a first major diameter **381** defined by the first helical thread **310**.
(101) In some embodiments, the distal shaft portion **309** may have a second minor diameter **322** and a second major diameter **382** defined by the second helical thread **320**.
(102) In some embodiments, at least a portion of the first minor diameter **321** and/or the second

minor diameter **322** may be constant. However, it will also be understood that in some embodiments at least a portion of the first minor diameter **321** and/or the second minor diameter **322** may not be constant.
(103) In some embodiments, at least a portion of the first major diameter **381** and/or the second major diameter **382** may be constant. However, it will also be understood that in some embodiments at least a portion of the first major diameter **381** and/or the second major diameter **382** may not be constant.
(104) In some embodiments, the second minor diameter **322** of the distal shaft portion **309** may be smaller than the first minor diameter **321** of the proximal shaft portion **308**.
(105) In some embodiments, the second major diameter **382** of the distal shaft portion **309** may be smaller than the first major diameter **381** of the proximal shaft portion **308**.
(106) In some embodiments, the first helical thread **310** may include a first concave undercut surface **331**, and the second helical thread **320** may include a second concave undercut surface **332**.
(107) In some embodiments, the first concave undercut surface **331** and the second concave undercut surface **332** may be angled towards the distal end **302** of the shaft **305**.
(108) However, it will also be understood that the bone implant **300** may include any thread configuration, feature, or morphology described or contemplated herein with respect to any fastener/implant to achieve optimal fixation within a given bone/tissue. For example, in some embodiments the first and second helical threads **310**, **320** may comprise standard or inverted threading, a "dual start" thread configuration, etc. Moreover, it will also be understood that the bone implant **300** may be utilized in conjunction with (or within) any system or procedure described or contemplated herein.
(109) In some embodiments, the flange component **380** may be located toward the proximal end of the shaft **305**.
(110) In some embodiments, the flange component **380** may include a bone-facing surface **386** and an implement-facing surface **384**.
(111) In some embodiments, the bone-facing surface **386** may comprise a convex surface.
(112) In some embodiments, the bone-facing surface **386** may comprise a partial spherical shape.
(113) In some embodiments, the implement-facing surface **384** may comprise a flat circular surface.
(114) In some embodiments, the flange component **380** may include one or more passageways formed therethrough (not shown), similar to the one or more passageways **420** shown in FIGS. **4**A-**4**D.
(115) In some embodiments, the one or more passageways may be configured to receive one or more bone screws therethrough (not shown) and/or promote bone in-growth within the one or more passageways during the healing process.
(116) In some embodiments, the attachment feature (e.g., such as the implant post **390**, or the implant recess **391** shown in FIG. **21**) may be located at the proximal end **301** of the shaft **305** and may be configured to removably secure an implement to the bone implant **300**.
(117) In some embodiments, the implant recess **391** (see FIG. **21**) may be configured to removably couple with an implement, such as the insert **700** shown in FIGS. **7**A and **7**B and discussed below in more detail.
(118) In some embodiments, the implant post **390** may be configured to removably couple with an implement, such as the articulating head **600** shown in FIGS. **6**A and **6**B and discussed below in more detail.
(119) In some embodiments, the implant post **390** may project proximally from the flange component **380**.
(120) In some embodiments, the implant post **390** may comprise a cylindrical shape.
(121) In some embodiments, the implant post **390** may comprise a tapered cylindrical shape or a partial conical shape.

(122) In some embodiments, the implant post **390** may comprise a morse taper.
(123) FIGS. **4**A-**4**D illustrate various views of a fastener, implant, shoulder joint implant, or bone implant **400**, according to another embodiment of the present disclosure. Specifically, FIG. **4**A is a front perspective view of the bone implant **400**, FIG. **4**B is a rear perspective view of the bone implant **400**, FIG. **4**C is a side view of the bone implant **400**, and FIG. **4**D is a cross-sectional side view of the bone implant **400** taken along the line C-C in FIG. **4**C.
(124) The bone implant **400** may generally include a shaft **405** having a proximal end **401**, a distal end **402**, a longitudinal axis **403**, a helical thread **410**, one or more self-tapping features **407**, a flange component **480**, an attachment feature or implant post **490**, and a torque connection interface **406** formed in/on the implant post **490**.
(125) In some embodiments, the shaft **405** may have a minor diameter **421** and a major diameter **481** defined by the helical thread **410**.
(126) In some embodiments, at least a portion of the minor diameter **421** may be constant. However, it will also be understood that in some embodiments at least a portion of the minor diameter **421** may not be constant.
(127) In some embodiments, at least a portion of the major diameter **481** may be constant. However, it will also be understood that in some embodiments at least a portion of the major diameter **481** may not be constant.
(128) In some embodiments, the helical thread **410** may include a concave undercut surface **431**.
(129) In some embodiments, the concave undercut surface **431** may be angled towards the distal end **402** of the shaft **405**.
(130) However, it will also be understood that the bone implant **400** may include any thread configuration, feature, or morphology described or contemplated herein with respect to any fastener/implant to achieve optimal fixation within a given bone/tissue. For example, in some embodiments the helical thread **410** may comprise standard or inverted threading, a “dual start” thread configuration, etc. Moreover, it will also be understood that the bone implant **400** may be utilized in conjunction with (or within) any system or procedure described or contemplated herein.
(131) In some embodiments, the flange component **480** may be located toward the proximal end of the shaft **405**.
(132) In some embodiments, the flange component **480** may include a bone-facing surface **482** and an implement-facing surface **484**.
(133) In some embodiments, the bone-facing surface **482** may comprise a convex surface.
(134) In some embodiments, the bone-facing surface **482** may comprise a partial spherical shape.
(135) In some embodiments, the implement-facing surface **484** may comprise a flat circular surface.
(136) In some embodiments, the flange component **480** may include one or more passageways **420** formed therethrough.
(137) In some embodiments, the one or more passageways **420** may be configured to receive one or more bone screws therethrough (not shown) and/or promote bone in-growth within the one or more passageways **420** during the healing process.
(138) In some embodiments, the attachment feature (e.g., such as the implant post **490**, or alternatively an implant recess (not shown) similar to the implant recess **590** shown in FIG. **20**) may be located at the proximal end **401** of the shaft **405** and may be configured to removably secure an implement to the bone implant **400**, such as the insert **700** shown in FIGS. **7**A and **7**B.
(139) In some embodiments, the implant post **490** may be configured to removably couple with an implement, such as the articulating head **600** shown in FIGS. **6**A and **6**B.
(140) In some embodiments, the implant post **490** may project proximally from the flange component **480**.
(141) In some embodiments, the implant post **490** may comprise a cylindrical shape.
(142) In some embodiments, the implant post **490** may comprise a tapered cylindrical shape or a

partial conical shape.
(143) In some embodiments, the implant post **490** may comprise a morse taper.
(144) FIGS. 5A-5E illustrate various views of a fastener, implant, shoulder joint implant, or bone implant **500**, according to another embodiment of the present disclosure. Specifically, FIG. 5A is a front perspective view of the bone implant **500**, FIG. 5B is a rear perspective view of the bone implant **500**, FIG. 5C is a bottom view of the bone implant **500**, FIG. 5D is a top view of the bone implant **500**, and FIG. 5E is a side view of the bone implant **500**.
(145) The bone implant **500** may generally include a shaft **505** having a proximal end **501**, a distal end **502**, a longitudinal axis **503**, a helical thread **510**, one or more self-tapping features **507**, a flange component **580**, an attachment feature or implant recess **590**, and a central longitudinal passageway **520**.
(146) In some embodiments, the shaft **505** may have a minor diameter **521** and a major diameter **581** defined by the helical thread **510**.
(147) In some embodiments, at least a portion of the minor diameter **521** may be constant. However, it will also be understood that in some embodiments at least a portion of the minor diameter **521** may not be constant.
(148) In some embodiments, at least a portion of the major diameter **581** may be constant. However, it will also be understood that in some embodiments at least a portion of the major diameter **581** may not be constant.
(149) In some embodiments, the helical thread **510** may include a concave undercut surface **531**.
(150) In some embodiments, the concave undercut surface **531** may be angled towards the distal end **502** of the shaft **505**.
(151) However, it will also be understood that the bone implant **500** may include any thread configuration, feature, or morphology described or contemplated herein with respect to any fastener/implant to achieve optimal fixation within a given bone/tissue. For example, in some embodiments the helical thread **510** may comprise standard or inverted threading, a “dual start” thread configuration, etc. Moreover, it will also be understood that the bone implant **500** may be utilized in conjunction with (or within) any system or procedure described or contemplated herein.
(152) In some embodiments, the helical thread **510** disposed about the shaft **505** may define a threaded shaft portion **515** along a length of the shaft **505**.
(153) In some embodiments, a ratio of the length of the threaded shaft portion **515** to the minor diameter **521** of the shaft **505** may be less than 1.50.
(154) In some embodiments, a ratio of the length of the threaded shaft portion **515** to the minor diameter **521** of the shaft **505** may be less than 1.25.
(155) In some embodiments, a ratio of the length of the threaded shaft portion **515** to the minor diameter **521** of the shaft **505** may be less than 1.10.
(156) In some embodiments, a ratio of the length of the threaded shaft portion **515** to the minor diameter **521** of the shaft **505** may be equal to 1.0.
(157) In some embodiments, a ratio of the length of the threaded shaft portion **515** to the minor diameter **521** of the shaft **505** may be less than 1.0.
(158) In some embodiments, the flange component **580** may be located toward the proximal end of the shaft **505**.
(159) In some embodiments, the flange component **580** may include a bone-facing surface **582** and an implement-facing surface **584**.
(160) In some embodiments, the bone-facing surface **582** and/or the implement-facing surface **584** may each comprise a flat circular surface. However, it will also be understood that in some embodiments the bone-facing surface **582** and/or the implement-facing surface **584** may comprise convex surfaces, concave surfaces, partial spherical surfaces, etc.
(161) In some embodiments, the flange component **580** may be integrally formed with the shaft **505**. However, it will also be understood that the flange component **580** may be coupled to the shaft

505 via any suitable method including, but not limited to, a morse taper, a set screw, a tab, a locking element, etc., (not shown), without departing from the spirit or scope of the present disclosure.
(162) In some embodiments, the attachment feature (e.g., such as the implant recess **590**, or alternatively the implant post **591** shown in FIG. **21**) may be located at the proximal end **501** of the shaft **505** and may be configured to removably secure an implement or articulating member to the bone implant **500**.
(163) In some embodiments, the implant recess **590** may be configured to removably couple with an articulating member, such as the insert **700** shown in FIGS. **7**A and **7**B.
(164) In some embodiments, the implant post **591** may be configured to removably couple with an articulating member, such as the articulating head **600** shown in FIGS. **6**A and **6**B.
(165) In some embodiments, the implant post **591** may project proximally from the flange component **580**.
(166) In some embodiments, the implant post **591** may comprise a cylindrical shape.
(167) In some embodiments, the implant post **591** may comprise a tapered cylindrical shape or a partial conical shape.
(168) In some embodiments, the implant post **591** may comprise a morse taper.
(169) FIGS. **6**A and **6**B illustrate a perspective view and a side view of an implement, shoulder joint implement, articulating member, or articulating head **600**, according to an embodiment of the present disclosure.
(170) The articulating head **600** may generally include a proximal end **601**, a distal end **602**, a convex articular surface **610**, a post recess **620**, a flange component recess **630**, and a beveled surface **640**.
(171) In some embodiments, the articulating head **600** may comprise a glenoid head prosthesis which may be utilized in a reverse shoulder arthroplasty system (e.g., see articulating head **600** shown in FIGS. **19** and **20**).
(172) In some embodiments, the articulating head **600** may comprise a humeral head prosthesis which may be utilized in an anatomic shoulder arthroplasty (e.g., see articulating head **600** shown in FIG. **21**).
(173) In some embodiments, the convex articular surface **610** may comprise a convex semi-spherical articular surface.
(174) In some embodiments, the convex articular surface **610** of the articulating head **600** may be received within, and/or articulate against, the concave articular surface **710** of the insert **700** shown in FIGS. **7**A and **7**B.
(175) In some embodiments, the post recess **620** formed in the articulating head **600** may be shaped and configured to removably couple with the implant posts of the bone implants previously discussed herein.
(176) In some embodiments, the post recess **620** may comprise a cylindrical shape.
(177) In some embodiments, the post recess **620** may comprise a tapered cylindrical shape or a partial conical shape.
(178) In some embodiments, the post recess **620** may comprise a morse taper.
(179) In some embodiments, the flange component recess **630** formed in the distal end **602** of the articulating head **600** may be shaped and configured to removably couple with the flange components of the bone implants previously discussed herein.
(180) FIGS. **7**A and **7**B illustrate bottom and top perspective views of an implement, shoulder joint implement, articulating member, or insert **700**, according to an embodiment of the present disclosure.
(181) The insert **700** may generally include a proximal end **701**, a distal end **702**, a concave articular surface **710**, an outer surface **730**, an implant-facing surface **782**, and an insert post **720**.
(182) In some embodiments, the insert **700** may comprise a glenoid insert prosthesis which may be

utilized in a reverse shoulder arthroplasty system (e.g., see insert **700** shown in FIGS. **19** and **20**).
(183) In some embodiments, the insert **700** may comprise a humeral insert prosthesis which may be utilized in an anatomic shoulder arthroplasty (e.g., see insert **700** shown in FIG. **21**).
(184) In some embodiments, the concave articular surface **710** may comprise a concave semi-spherical articular surface.
(185) In some embodiments, the concave articular surface **710** of the insert **700** may receive, and/or articulate against, the convex articular surface **610** of the articulating head **600** shown in FIGS. **6**A and **6**B.
(186) In some embodiments, the insert post **720** of the insert **700** may be shaped and configured to removably couple with the implant recesses of the bone implants previously discussed herein.
(187) In some embodiments, the insert post **720** may comprise a cylindrical shape.
(188) In some embodiments, the insert post **720** may comprise a tapered cylindrical shape or a partial conical shape.
(189) In some embodiments, the insert post **720** may comprise a morse taper.
(190) FIGS. **8-19** illustrate an example shoulder arthroplasty procedure, according to an embodiment of the present disclosure. Specifically, FIG. **8** is a front view of a shoulder joint **800** comprising a scapula bone **810** and a humeral bone **850**, FIGS. **9-12** illustrate preparation of the scapula bone **810**, FIGS. **13-16** illustrate preparation of the humeral bone **850**, FIG. **17** is a front view of the prepared shoulder joint **800**, FIG. **18** is a front view of the shoulder joint **800** with a reverse shoulder arthroplasty system installed therein, and FIG. **19** is a cross-sectional side view of FIG. **18**.
(191) FIGS. **9-12** illustrate preparation of the scapula bone **810** during the example shoulder arthroplasty procedure. Specifically, FIG. **9** is a perspective view of the scapula bone **810** before preparation, FIG. **10** shows the scapula bone of FIG. **9** after glenoid preparation, FIG. **11** is a side view of the prepared scapula bone, and FIG. **12** shows a cross-sectional side view of the scapula bone of FIG. **11** taken along the line D-D in FIG. **11**.
(192) In some embodiments, the glenoid cavity **815** may be prepared by reaming a surface of the glenoid cavity **815** with a suitable reamer tool (not shown) to form a prepared glenoid surface **816**.
(193) In some embodiments, the prepared glenoid surface **816** may comprise a substantially flat surface.
(194) In some embodiments, the prepared glenoid surface **816** may comprise a concave surface.
(195) In some embodiments, the prepared glenoid surface **816** may comprise a concave semi-spherical surface.
(196) In some embodiments, the glenoid cavity **815** may be further prepared by forming a glenoid bone tunnel **820** within the scapula bone **810** with one or more drill tools (not shown).
(197) In some embodiments, the glenoid bone tunnel **820** may comprise a proximal bone tunnel portion **821** and a distal bone tunnel portion **822**.
(198) In some embodiments, a diameter of the proximal bone tunnel portion **821** may be sized and shaped to receive the first minor diameter **321** of the bone implant **300** therein (see FIG. **3**D), and a diameter of the distal bone tunnel portion **822** may be sized and shaped to receive the second minor diameter **322** of the bone implant **300** therein.
(199) In some embodiments, the diameter of the proximal bone tunnel portion **821** may be smaller than the first minor diameter **321** of the bone implant **300**, and/or the diameter of the distal bone tunnel portion **822** may be smaller than the second minor diameter **322** of the bone implant **300**.
(200) In some embodiments, the diameter of the proximal bone tunnel portion **821** may be equal to the first minor diameter **321** of the bone implant **300**, and/or the diameter of the distal bone tunnel portion **822** may be equal to the second minor diameter **322** of the bone implant **300**.
(201) In some embodiments, the diameter of the proximal bone tunnel portion **821** may be larger than the first minor diameter **321** of the bone implant **300**, and/or the diameter of the distal bone tunnel portion **822** may be larger than the second minor diameter **322** of the bone implant **300**.

(202) In some embodiments, the proximal bone tunnel portion **821** may be tapped with a first bone tap tool (not shown) to form a first bone thread **831** about the proximal bone tunnel portion **821**.
(203) In some embodiments, the distal bone tunnel portion **822** may be tapped with a second bone tap tool (not shown) to form a second bone thread **832** about the distal bone tunnel portion **822**.
(204) It will be understood that the first bone thread **831** and/or the second bone thread **832** may be sized and shaped to receive any thread configuration, feature, or morphology described or contemplated herein with respect to any fastener/implant to achieve optimal fixation within a given bone/tissue.
(205) FIGS. **13-16** illustrate preparation of the humeral bone **850** during the example shoulder arthroplasty procedure. Specifically, FIG. **13** is a perspective view of the humeral bone **850** before preparation, FIG. **14** shows the humeral bone **850** with a prepared humeral head **855**, FIG. **15** is a side view of the prepared humeral bone, and FIG. **16** is a cross-sectional side view of FIG. **15**.
(206) In some embodiments, the humeral head **855** may be prepared by cutting and/or reaming the humeral head **855** with a suitable cutter and/or reamer tool (not shown) to form a prepared humeral head surface **856**.
(207) In some embodiments, the prepared humeral head surface **856** may comprise a substantially flat surface.
(208) In some embodiments, the prepared humeral head surface **856** may comprise a concave surface.
(209) In some embodiments, the prepared humeral head surface **856** may comprise a concave semi-spherical surface.
(210) In some embodiments, the humeral head **855** may be further prepared by forming a humeral bone tunnel **860** within the humeral head **855** with a drill tool (not shown).
(211) In some embodiments, a diameter of the humeral bone tunnel **860** may be sized and shaped to receive the minor diameter **521** of the bone implant **500** (see FIG. **5**E).
(212) In some embodiments, the diameter of the humeral bone tunnel **860** may be smaller than the minor diameter **521** of the bone implant **500**.
(213) In some embodiments, the diameter of the humeral bone tunnel **860** may be equal to the minor diameter **521** of the bone implant **500**.
(214) In some embodiments, the diameter of the humeral bone tunnel **860** may be larger than the minor diameter **521** of the bone implant **500**.
(215) In some embodiments, the humeral bone tunnel **860** may be tapped with a bone tap tool (not shown) to form a bone thread **861** about the humeral bone tunnel **860**.
(216) It will be understood that the bone thread **861** may be sized and shaped to receive any thread configuration, feature, or morphology described or contemplated herein with respect to any fastener/implant to achieve optimal fixation within a given bone/tissue.
(217) FIG. **17** shows the prepared shoulder joint bones before implant installation. FIG. **18** shows the prepared shoulder joint of FIG. **17** with the reverse shoulder arthroplasty system of FIG. **20** installed therein, and FIG. **19** shows a cross-sectional side view of FIG. **18**. Alternatively, the anatomic shoulder arthroplasty system shown in FIG. **21** may be installed in the prepared shoulder joint of FIG. **17**.
(218) As noted above, FIGS. **20** and **21** illustrate front perspective views of the reverse shoulder arthroplasty system and the anatomic shoulder arthroplasty system, respectively. The reverse shoulder arthroplasty system shown in FIG. **20** may utilize the components shown and discussed in FIGS. **3**A-**3**D and **5**A-**7**B and installed in the shoulder joint **800** illustrated in FIG. **19**. In this manner, the implant recess **590** of the bone implant **500** may receive the insert post **720** of the insert **700** therein, and the post recess **620** of the articulating head **600** may receive the implant post **390** of the bone implant **300** therein for the reverse shoulder arthroplasty system of FIG. **20**.
(219) Alternatively, the anatomic shoulder arthroplasty system shown in FIG. **21** may utilize components with complementary features to that of the reverse shoulder arthroplasty system shown

in FIG. **20**. For example, the post recess **620** of the articulating head **600** may receive the implant post **591** of the bone implant **500** therein, and the implant recess **391** of the bone implant **300** may receive the insert post **720** of the insert **700** therein for the anatomic shoulder arthroplasty system shown in FIG. **21**.

(220) FIGS. **22**A-**22**D illustrate various views of a fastener, implant, or bone implant **900**, according to another embodiment of the present disclosure. Specifically, FIG. **22**A is a front perspective view of the bone implant **900**, FIG. **22**B is a rear perspective view of the bone implant **900**, FIG. **22**C is a side view of the bone implant **900**, and FIG. **22**D is a cross-sectional side view of the bone implant **900** taken along the line E-E in FIG. **22**C.

(221) The bone implant **900** may generally include a tapered shaft **905** having a proximal end **901**, a distal end **902**, a longitudinal axis **903**, at least one tapered helical thread **910** disposed about the tapered shaft, and a torque connection interface **906** formed in/on the proximal end **901** of the tapered shaft **905**.

(222) In some embodiments, the distal end **902** of the tapered shaft **905** may comprise a pointed or sharp tip.

(223) In some embodiments, the tapered shaft **905** may comprise one or more self-tapping features or cutting flutes (not shown).

(224) In some embodiments, the tapered shaft **905** may have a continuously variable minor diameter **921** generally defined by the shape of the tapered shaft **905**, and a continuously variable major diameter **981** generally defined by the shape of the at least one tapered helical thread **910** disposed about the tapered shaft **905**.

(225) In some embodiments, the continuously variable minor diameter **921** defined by the shape of the tapered shaft **905** may comprise an at least partially conical shape.

(226) In some embodiments, the continuously variable major diameter **981** defined by the shape of the at least one tapered helical thread **910** disposed about the tapered shaft **905** may comprise an at least partially conical shape.

(227) In some embodiments, the continuously variable minor diameter **921** defined by the shape of the tapered shaft **905** may generally decrease moving from the proximal end **901** of the tapered shaft **905** toward the distal end **902** of the tapered shaft **905**.

(228) In some embodiments, the continuously variable major diameter **981** defined by the shape of the at least one tapered helical thread **910** disposed about the tapered shaft **905** may generally decrease moving from the proximal end **901** of the tapered shaft **905** toward the distal end **902** of the tapered shaft **905**.

(229) In some embodiments, the at least one tapered helical thread **910** may include at least one concave undercut surface **931**.

(230) In some embodiments, the at least one concave undercut surface **931** may be angled towards one of the proximal end **901** and the distal end **902** of the tapered shaft **905**.

(231) However, it will also be understood that the bone implant **900** may include any thread configuration, feature, or morphology described or contemplated herein with respect to any fastener/implant to achieve optimal fixation within a given bone/tissue. For example, in some embodiments the at least one tapered helical thread **910** may comprise standard or inverted threading, a “dual start” thread configuration, crescent shapes, etc.

(232) Moreover, it will also be understood that the bone implant **900**, or portions of the general design/shape of the bone implant **900**, may be utilized in conjunction with (or within) any fastener, bone implant, system, or procedure that is described or contemplated herein. For example, any of the fasteners/bone implants described or contemplated herein may generally be configured to include a tapered shaft with tapered helical threading disposed about the tapered shaft. As one such non-limiting example, the design of the bone implant **300** shown in FIGS. **3**A-**3**D may be modified to include a tapered shaft with tapered helical threading disposed about the tapered shaft in place of, or in addition to, the proximal shaft portion **308** and/or the distal shaft portion **309** previously

discussed with reference to FIGS. **3**A-**3**D, etc.

(233) Any of the implants described or contemplated herein may be configured for removal and replacement during a revision procedure by simply unscrewing and removing the implant from the bone/tissue in which the implant resides. Moreover, the implants described herein may advantageously be removed from bone without removing any appreciable amount of bone during the removal process to preserve the bone. In this manner, implants may be mechanically integrated with the bone, while not being cemented to the bone or integrated via bony ingrowth, in order to provide an instant and removable connection between an implant and a bone. Accordingly, revision procedures utilizing the implants described herein can result in less trauma to the bone and improved patient outcomes. However, it will also be understood that any of the implants described or contemplated herein may also be utilized with cement, as desired.

(234) Any procedures/methods disclosed herein comprise one or more steps or actions for performing the described method. The method steps and/or actions may be interchanged with one another. In other words, unless a specific order of steps or actions is required for proper operation of the embodiment, the order and/or use of specific steps and/or actions may be modified.

(235) Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure, or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

(236) Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the present disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any embodiment requires more features than those expressly recited in that embodiment. Rather, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment.

(237) Recitation of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112(f). It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles set forth herein.

(238) The phrases “connected to,” “coupled to” and “in communication with” refer to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be functionally coupled to each other even though they are not in direct contact with each other. The term “coupled” can include components that are coupled to each other via integral formation, as well as components that are removably and/or non-removably coupled with each other. The term “abutting” refers to items that may be in direct physical contact with each other, although the items may not necessarily be attached together. The phrase “fluid communication” refers to two or more features that are connected such that a fluid within one feature is able to pass into another feature. Moreover, as defined herein the term “substantially” means within +/−20% of a target value, measurement, or desired characteristic.

(239) While specific embodiments and applications of the present disclosure have been illustrated and described, it is to be understood that the scope of this disclosure is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the devices, systems, and methods disclosed herein.

Claims

1. A bone implant comprising: a shaft comprising: a proximal end; a distal end; a longitudinal axis; a proximal shaft portion comprising: a first minor diameter; and a first helical thread disposed about the proximal shaft portion defining a first major diameter of the proximal shaft portion, the first helical thread comprising a first concave undercut surface; and a distal shaft portion comprising: a second minor diameter; and a second helical thread disposed about the distal shaft portion defining a second major diameter of the distal shaft portion, the second helical thread comprising a second concave undercut surface, wherein: the first concave undercut surface and the second concave undercut surface are angled towards one of the proximal end and the distal end of the shaft; the first concave undercut surface extends from the first minor diameter of the proximal shaft portion to a first crest of the first helical thread disposed about the proximal shaft portion; the second concave undercut surface extends from the second minor diameter of the distal shaft portion to a second crest of the second helical thread disposed about the distal shaft portion; the first minor diameter of the proximal shaft portion is constant; the second minor diameter of the distal shaft portion is constant; the second minor diameter of the distal shaft portion is smaller than the first minor diameter of the proximal shaft portion; and the second major diameter of the distal shaft portion is smaller than the first major diameter of the proximal shaft portion.
2. The bone implant of claim 1, comprising a flange component at the proximal end of the shaft, the flange component comprising: a bone-facing surface; and an implement-facing surface.
3. The bone implant of claim 2, wherein the bone-facing surface comprises a convex surface.
4. The bone implant of claim 2, wherein the bone-facing surface comprises a semi-spherical surface.
5. The bone implant of claim 1, comprising an attachment feature at the proximal end of the shaft configured to secure an implement.
6. The bone implant of claim 5, wherein: the attachment feature comprises a post; and the implement comprises an articulating head comprising a convex semi-spherical articular surface, wherein the articulating head is configured to be removably couplable with the post.
7. The bone implant of claim 5, wherein: the attachment feature comprises an implant recess; and the implement comprises an insert having a concave semi-spherical articular surface, wherein the insert is configured to be removably couplable with the implant recess.
8. A bone implant comprising: a shaft comprising: a proximal end; a distal end; a longitudinal axis; a minor diameter; and a threaded shaft portion; an articulating member disposed at the proximal end of the shaft; and a helical thread disposed about the shaft defining a length of the threaded shaft portion, the helical thread comprising a concave undercut surface on a first side of the helical thread, and a convex undercut surface on a second side of the helical thread, wherein: the concave undercut surface extends from the minor diameter of the shaft to a crest of the helical thread on the first side of the helical thread; the convex undercut surface extends from the minor diameter of the shaft to the crest of the helical thread on the second side of the helical thread; the concave undercut surface is angled towards one of the proximal end and the distal end of the shaft; and a ratio of the length of the threaded shaft portion to the minor diameter of the shaft is less than 1.50.
9. The bone implant of claim 8, wherein the ratio of the length of the threaded shaft portion to the minor diameter of the shaft is less than 1.25.
10. The bone implant of claim 8, wherein the ratio of the length of the threaded shaft portion to the minor diameter of the shaft is less than 1.10.
11. The bone implant of claim 8, wherein the ratio of the length of the threaded shaft portion to the minor diameter of the shaft is equal to 1.0.
12. The bone implant of claim 8, wherein the ratio of the length of the threaded shaft portion to the minor diameter of the shaft is less than 1.0.
13. The bone implant of claim 8, further comprising: a flange component at the proximal end of the shaft, the flange component comprising: a bone-facing surface; and an implement-facing surface.

14. The bone implant of claim 8, wherein the articulating member is configured to be removably couplable with an attachment feature at the proximal end of the shaft.
15. A shoulder joint implant comprising: a shaft comprising: a proximal end; a distal end; and a longitudinal axis; a helical thread disposed about the shaft along the longitudinal axis between the proximal and distal ends of the shaft, the helical thread comprising: a first undercut surface; a second undercut surface; a third undercut surface; and a fourth open surface, wherein: the helical thread does not have mirror symmetry with itself across any plane perpendicular to the longitudinal axis of the shaft; at least one of the first undercut surface, the second undercut surface, the third undercut surface, and the fourth open surface comprises a flat surface; the first undercut surface and the third undercut surface are angled towards one of the proximal end and the distal end of the shaft; and the second undercut surface and the fourth open surface are angled towards the other one of the proximal end and the distal end of the shaft; and a shoulder joint implement comprising an articular surface at the proximal end of the shaft.
16. The shoulder joint implant of claim 15, wherein: the shoulder joint implement comprises a glenoid head prosthesis; and the articular surface comprises a convex semi-spherical articular surface.
17. The shoulder joint implant of claim 15, wherein: the shoulder joint implement comprises a humeral head prosthesis; and the articular surface comprises a convex semi-spherical articular surface.
18. The shoulder joint implant of claim 15, wherein: the shoulder joint implement comprises a glenoid insert prosthesis; and and the articular surface comprises a concave semi-spherical articular surface.
19. The shoulder joint implant of claim 15, wherein: the shoulder joint implement comprises a humeral insert prosthesis; and the articular surface comprises a concave semi-spherical articular surface.
20. The shoulder joint implant of claim 15, comprising a flange component at the proximal end of the shaft, the flange component comprising: a bone-facing surface; and an implement-facing surface.
21. The bone implant of claim 1, wherein the second major diameter of the distal shaft portion is smaller than the first minor diameter of the proximal shaft portion.
22. The bone implant of claim 8, wherein the minor diameter of the shaft is constant.
23. The shoulder joint implant of claim 15, wherein: the first undercut surface and the second undercut surface form a concave undercut surface on a first side of the helical thread; and the third undercut surface and the fourth open surface form a convex undercut surface on a second side of the helical thread.
24. A bone implant comprising: a shaft comprising: a proximal end; a distal end; a longitudinal axis; a proximal shaft portion comprising: a first minor diameter; and a first helical thread disposed about the proximal shaft portion defining a first major diameter of the proximal shaft portion, the first helical thread comprising a first concave undercut surface; and a distal shaft portion comprising: a second minor diameter; and a second helical thread disposed about the distal shaft portion defining a second major diameter of the distal shaft portion, the second helical thread comprising a second concave undercut surface, wherein: the first helical thread does not have mirror symmetry with itself across any plane perpendicular to the longitudinal axis of the shaft; the second helical thread does not have mirror symmetry with itself across any plane perpendicular to the longitudinal axis of the shaft; the first concave undercut surface and the second concave undercut surface are angled towards one of the proximal end and the distal end of the shaft; the second minor diameter of the distal shaft portion is smaller than the first minor diameter of the proximal shaft portion forming a shoulder intermediate the proximal shaft portion and the distal shaft portion; and the second major diameter of the distal shaft portion is smaller than the first major diameter of the proximal shaft portion.

25. The bone implant of claim 24, wherein the second major diameter of the distal shaft portion is smaller than the first minor diameter of the proximal shaft portion.
26. The bone implant of claim 24, wherein: the first minor diameter of the proximal shaft portion is constant; and the second minor diameter of the distal shaft portion is constant.
27. A bone implant comprising: a shaft comprising: a proximal end; a distal end; a longitudinal axis; a minor diameter; and a threaded shaft portion; an articulating member disposed at the proximal end of the shaft; and a helical thread disposed about the shaft defining a length of the threaded shaft portion, the helical thread comprising a concave undercut surface, wherein: the concave undercut surface is angled towards one of the proximal end and the distal end of the shaft; and the concave undercut surface is formed by a first undercut surface, a second undercut surface, and an inflection point intermediate the first undercut surface and the second undercut surface, wherein the first undercut surface extends from the minor diameter of the shaft to the inflection point, and the second undercut surface extends from the inflection point to a crest of the helical thread.
28. The bone implant of claim 27, wherein the minor diameter of the shaft is constant.
29. The bone implant of claim 27, wherein: the concave undercut surface is positioned on a first side of the helical thread; and the helical thread comprises a convex undercut surface positioned on a second side of the helical thread.
30. A shoulder joint implant comprising: a shaft comprising: a proximal end; a distal end; and a longitudinal axis; a helical thread disposed about the shaft along the longitudinal axis between the proximal and distal ends of the shaft, the helical thread comprising: a first undercut surface; a second undercut surface adjacent the first undercut surface forming a first inflection point intermediate the first undercut surface and the second undercut surface; a third undercut surface; and a fourth open surface adjacent the third undercut surface forming a second inflection point intermediate the third undercut surface and the fourth open surface, wherein: the first undercut surface extends from a minor diameter of the shaft to the first inflection point, and the second undercut surface extends from the first inflection point to a crest of the helical thread; the third undercut surface extends from the minor diameter of the shaft to the second inflection point, and the fourth open surface extends from the second inflection point to the crest of the helical thread; the first undercut surface and the third undercut surface are angled towards one of the proximal end and the distal end of the shaft; and the second undercut surface and the fourth open surface are angled towards the other one of the proximal end and the distal end of the shaft; and a shoulder joint implement comprising an articular surface at the proximal end of the shaft.
31. The shoulder joint implant of claim 30, wherein: the first undercut surface and the second undercut surface form a concave undercut surface on a first side of the helical thread; and the third undercut surface and the fourth open surface form a convex undercut surface on a second side of the helical thread.
32. The shoulder joint implant of claim 30, wherein at least one of the first undercut surface, the second undercut surface, the third undercut surface, and the fourth open surface comprises a flat surface.